

Spaced practice and long-term recall

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Abstract

We compare spaced and massed practice in 120 first-year students learning the same biology vocabulary. Students who spread four short sessions over two weeks recalled 71% of the terms after 30 days, against 38% for students who studied in one block. Figure 1 shows recall over time and Table 1 gives the scores. Section 2 describes the method. The forgetting model follows earlier work [1].

1 Introduction

Memory fades quickly after a single study session. A common description is the forgetting curve, where the share of material retained after t days follows $R(t) = e^{(-t/S)}$. Here S is the stability of the memory: the larger S , the slower the decline. Equation (1) on the next page gives the fitted form.

Reviewing material at growing intervals is thought to raise S with each session, so later reviews can be further apart. This spacing effect is well documented in the laboratory [2], but less is known about its size in an ordinary course with real deadlines.

1.1 Research question

Does spreading the same total study time over two weeks improve recall one month later, and by how much? We also ask whether the benefit holds for students with lower prior grades.

2 Method and results

Students were randomly assigned to a spaced group (four 20-minute sessions on days 0, 2, 6 and 13) or a massed group (one 80-minute session on day 0). Both groups took the same recall test on days 1, 7, 14 and 30.

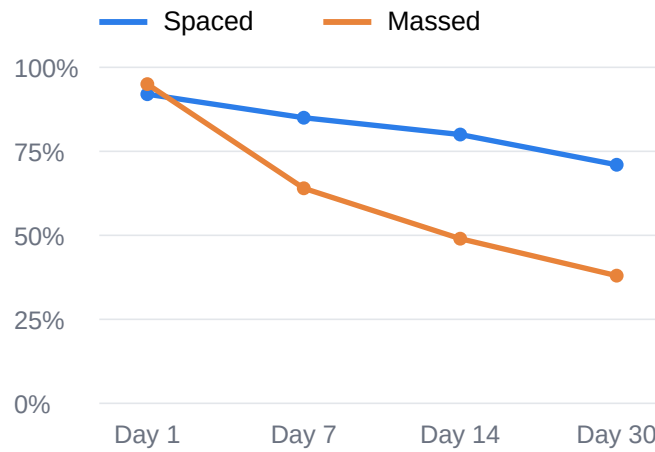


Figure 1: Share of terms recalled by each group, from day 1 to day 30.

Table 1: Mean recall (%) by test day.

Test day	Spaced	Massed	Difference
1	92	95	−3
7	85	64	+21
14	80	49	+31
30	71	38	+33

$$R(t) = e^{(-t/S)} \quad (1)$$

Equation (1) is fitted to each group's scores by least squares. Stability S is 86 days for the spaced group and 31 days for the massed group, so spaced practice slowed forgetting by a factor of about 2.8.

3 Participant consent form

Please complete this form before your first session. Fields marked * are required.

Full name *

Date of birth (DD/MM/YYYY) *

Email *

Phone

Faculty

Science 

Preferred session

☐

Morning

☐

Afternoon

I agree to take part *

☐

Signature *

Date *

References

[1] R. Alvarez and J. Singh. Modelling forgetting in classroom settings. Journal of Learning Research, 8(2):101-118, 2022.

[2] M. Okafor. Spacing effects in the laboratory: a review. Memory Studies, 14:1-29, 2019.